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Selmax technology & camelid phage display for the discovery of anti-SARS-CoV-2 antibodies

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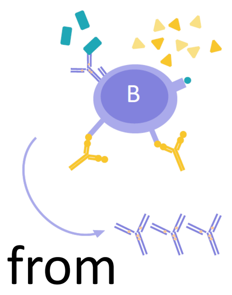
Introduction

The ongoing COVID-19 pandemic situation caused by SARS-CoV-2 and variants of concern such as B.1.1.529 (Omicron) is posing multiple challenges to humanity. The rapid evolution of the virus requires adaptation of diagnostic and therapeutic applications.

Generation of human anti-Spike antibodies

In vitro immunization



Human PBMC were isolated from Buffy coat and stimulated with target. Increasing yields of target specific antibodies of subclass G after *in vitro* immunization were detected. B-lymphocytes were fused to transgenic myeloma cell line.

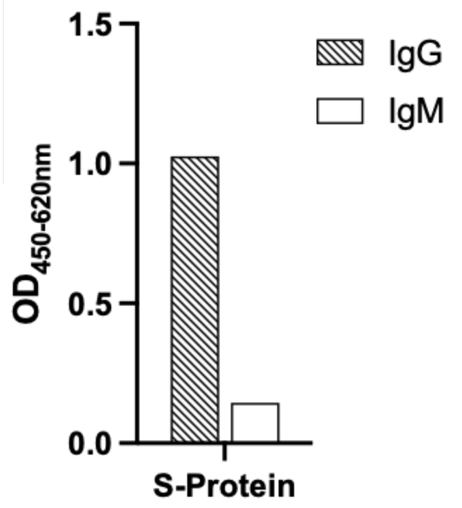
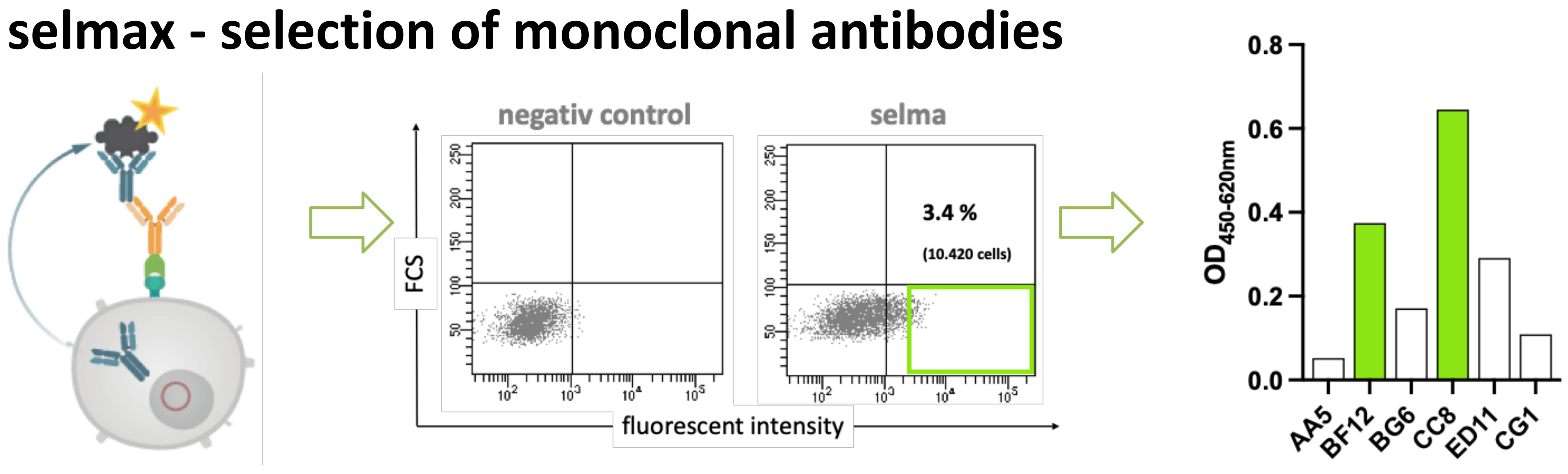


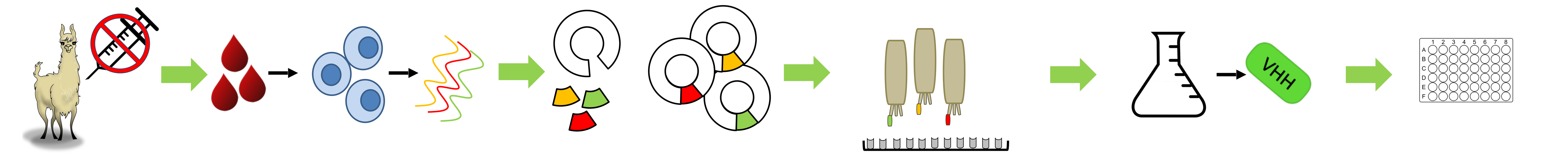
Fig. 1: Indirect ELISA to confirm successful in vitro immunization.

selmax - selection of monoclonal antibodies



Target-specific selection of antibody producing hybridoma cells were isolated via FACS. In a final ELISA target specificity was confirmed.

Generation of camelid anti-Spike antibodies



Prokaryotic expression and purification of VHH B10

Following overnight expression, the periplasmic fraction was isolated and purified via Ni-NTA (Fig. 2A). Elution 1 was then used for further purification by size exclusion chromatography (Fig. 2B). SDS-Page was performed after each purification step. The result of the purification process was a highly purified VHH.

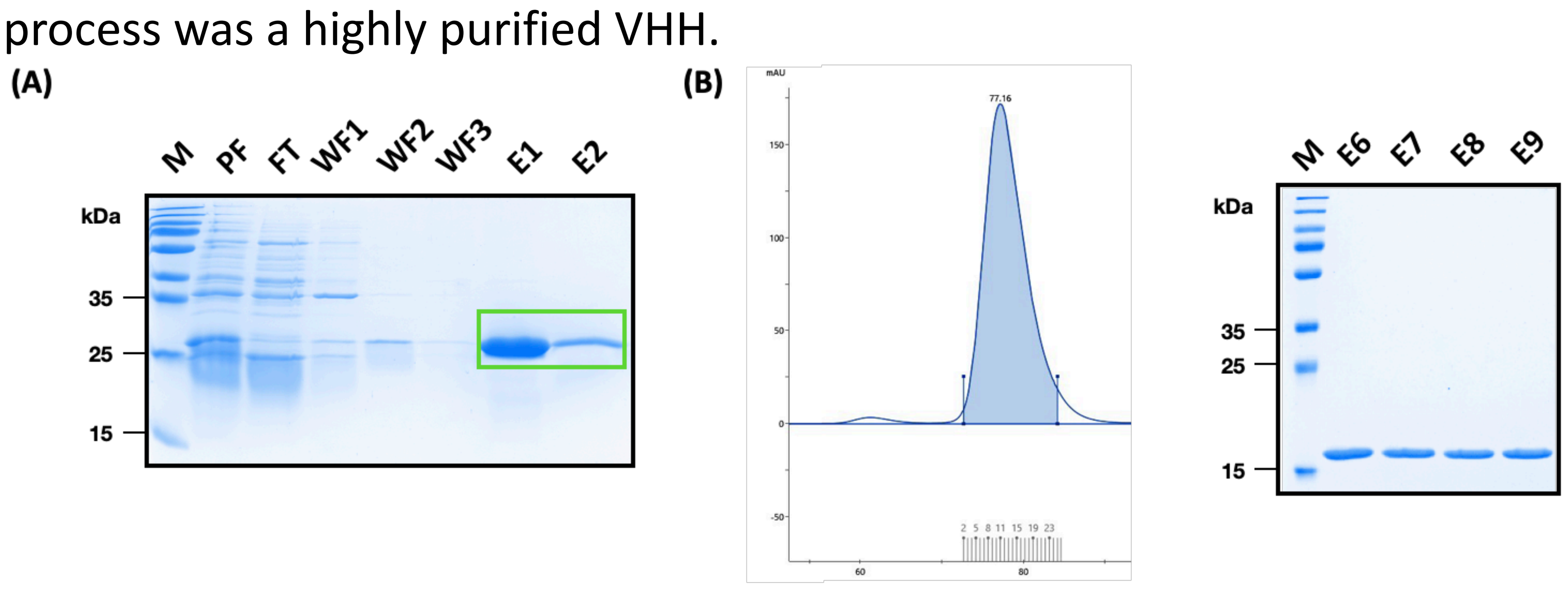


Fig. 2: Two step purification process of VHH B10 via (A) Ni-NTA (B) size exclusion chromatographie. Protein fraction in Coomassie-stained 15 % PAA gel under reducing conditions. VHH B10 with calculated 15 kDa. PF - Periplasmic fraction, FT - Flow Through, WF - Wash fraction, E - Eluate fraction, M - Protein standard PageRuler Prestained Plus (ThermoFisher Scientific).

hcAb-upgrade for VHH B10

The VHH B10 was cloned into our in-house vector pNEM_hcAb. This results in a VHH with a camelid IgG2 Fc domain (= hcAb). After transient transfection into Expi293 and cultivation for 7 days, the supernatant was harvested. Subsequently, hcAb B10 was purified via protein A.

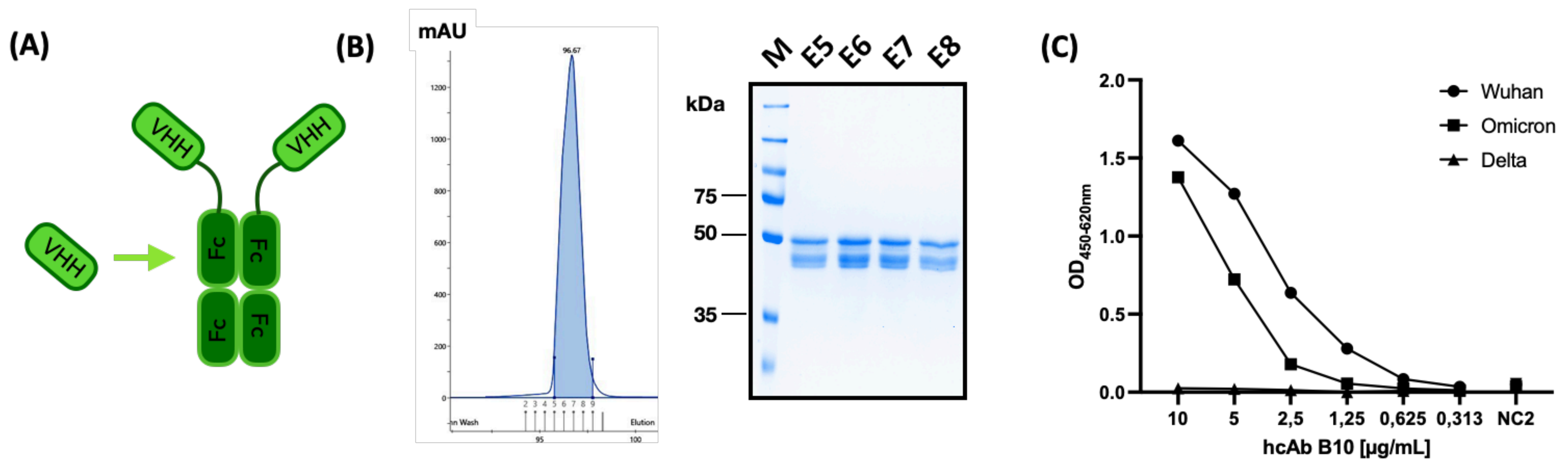


Fig. Z: (A) Schematic illustration of VHH updated to hcAb (B) Affinity purification via protein A (ÄKTA). Elution fraction (E) 5-8 (fraction 4, 14-15 not shown) in Coomassie-stained 10 % PAA gel under reducing conditions. HcAb B10 with calculated 42,5 kDa as monomer, non-glycosylated. PF - Periplasmic fraction, FT - Flow Through, WF - Wash fraction, E - Eluate fraction, M - Protein standard PageRuler Prestained Plus (ThermoFisher Scientific).(C) Coated spike protein was recognized by hcAb B10 in an indirect ELISA. The spike protein of Wuhan and Omicron could be detected by the hcAb. No signals were detected for Delta sublinage. In-house anti-camelid IgG2 antibody used as detector.

VHH B10 binds specific Wuhan and Omicron spike protein

VHH B10 recognizes its target spike protein (Wuhan) and also the sublineage Omicron (B.1.1.529). It does not recognize Delta (B.1.617.2) in an indirect ELISA.

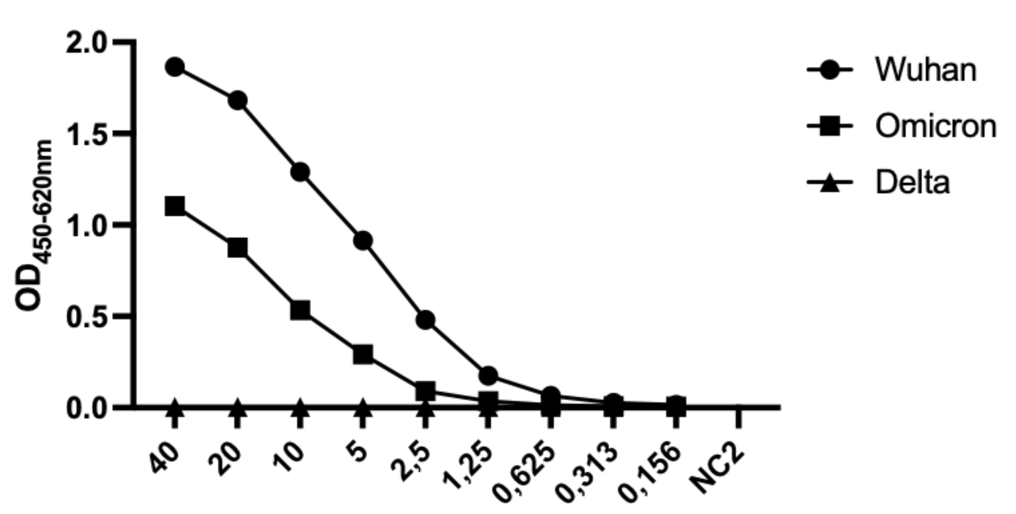


Fig. Y: Coated spike protein was recognized by VHH B10 after purification in an indirect ELISA. The spike protein of Wuhan and Omicron could be detected by the VHH. No signals were detected for Delta sublinage. Detection of VHH by anti-His POD (Sigma).

Affinity measurements with Monolith

The calculated Kd for binding of hcAb B10 to Wuhan was 0.92 nM ± 0.69 and for the Omicron trimer was 9.03 nM ± 0.30.

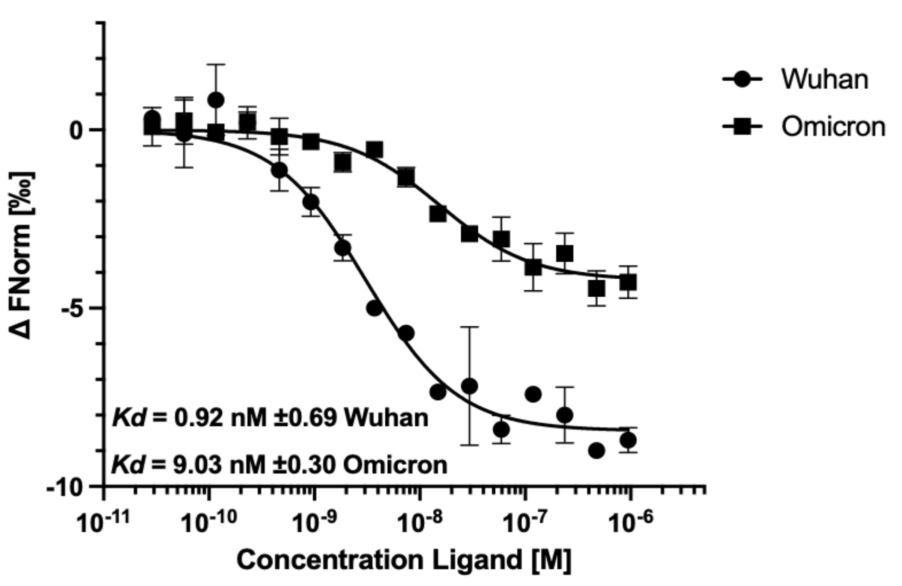


Fig. Y: Binding affinity analysis with a fixed concentration of NT-647 conjugated hcAb B10 (20 nM) and variable concentrations of Wuhan or Omicron as unlabeled binding partner varied from 0.95 µM to 2.9E-05 µM. An MST on time of 1.5 s was used for analysis. For Kd determination, the resulting dose-response curves were fitted to a single-site binding model. MST measurements were performed in duplicate (n=2) for Wuhan and in triplicate (n=3) for Omicron. Error bars represent standard deviation.

References:

[1] Listek, M., Micheel, B., Hanack, K. (2015) EP3134435A1 „Biomolecule releasing cell and selection thereof by means of a surface protein" patent owner: new/era/mabs GmbH

[2] Listek, M., Hönow, A., Gossen, M., Hanack, K. (2020) „A novel selection strategy for antibody producing hybridoma cells based on a new transgenic fusion cell line" Sci Rep. 10(1):1664

[3] Schlör, A. Hirschberg, S., Amor, GB., Meister, TL., Arora, P., Pöhlmann, S., Hoffmann, M., Pfaender, S., Eddin, OK., Kamhieh-Milz, J. and Hanack, K. (2022) "SARS-CoV-2 neutralizing camelid heavy-chain-only antibodies as powerful tools for diagnostic and therapeutic applications" Front. Immunol. 13:930975.

Acknowledgement:

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